Alexis DHERMY

Applicant for M.S. Program in Artificial Intelligence – South Korea (Spring 2027)

2003.02.12 | French | +33 7 51 69 66 84 | a.dhermy@live.com | LinkedIn: alexis-dhermy | Website: alexis-dhermy.github.io

Education

EPF Graduate School of Engineering — Paris-Cachan, France Sep 2021 – Aug 2026
Master of Science in Engineering (French Diplôme d'Ingénieur)
Major: Intelligent Systems & Digital Technologies | Track: Artificial Intelligence & Cloud Computing

- Curriculum: 5-year integrated program | 2-year intensive preparatory cycle (Math & Physics) followed by a 3-year engineering cycle (1-year generalist core + 2-year CS & AI specialization)
- GPA: **16.09/20** | Rank: **Top 3% of cohort** (Y4: 16/276, Y3: 6/206, Y2: 4/178, Y1: 4/217)
- Thesis: LLM-Based RAG Architecture for Enterprise Systems (*Viridien*) | Advisor: Prof. Zehira Haddad
- Relevant Coursework: Deep Learning, Machine Learning, Generative AI, Calculus, Linear Algebra, Probability & Statistics

Sungkyunkwan University — Seoul, South Korea Sep 2024 – Dec 2024
Exchange Semester
College: Computing & Informatics | Department: Applied Artificial Intelligence

- GPA: **4.10/4.5**
- Relevant Coursework: Introduction to Artificial Intelligence, Introduction to Reinforcement Learning, Culture Data and Machine Learning, Language Technology and Cultural Content

Honors & Awards

National Engineering Entrance Merit Distinction — Concours Avenir (*Grand Classé*) May 2021

- Selected in the top tier of 10,000+ candidates for academic excellence, in one of France's most competitive post-high school engineering entrance exams, earning full written test exemption.

Research Interests

- Natural Language Processing (NLP), Large Language Models (LLMs), AI Agents, Information Retrieval
- Reasoning Models, Reinforcement Learning, Model Evaluation, Efficient Inference

Academic Projects

Music Recommendation Using NLP [Code & Report] — Sungkyunkwan University Nov 2024 – Dec 2024

- Developed an end-to-end NLP pipeline (NLTK, SpaCy, BERT) to perform tokenization, POS tagging, lemmatization, and sentiment analysis on top Spotify tracks across France, Spain, and Korea.
- Built a cross-cultural recommendation engine that maps semantic and emotional features to recommend French/Spanish tracks based on a target Korean song.

Teaching a Robot to Walk with RL [Presentation] — Sungkyunkwan University Nov 2024 – Dec 2024

- Trained a 2D bipedal robot in OpenAI Gym (BipedalWalker-v3) to navigate complex terrain through deep reinforcement learning.
- Designed custom reward shaping combining speed, distance, and joint stability with a step-based delay threshold, driving a 3x increase in distance and velocity over baseline Advantage Actor-Critic (A2C).
- Implemented Proximal Policy Optimization (PPO) to mitigate policy collapse, boosting average episode moving distance by 132%.

Professional Experience

AI Software Engineer — Viridien (R&D Department – Seismic Imaging), Massy, France Feb 2026 – Jul 2026
Master's Thesis Internship

- Conducted applied AI research to architect an enterprise advanced RAG system (LlamaIndex, Python), parsing 80k+ geophysical technical documents and DSL scripts to boost response accuracy from 54% to 90%.
- Built a multi-stage ingestion pipeline featuring Docling parsing, automated cleaning, hierarchical chunking, Anthropic's Contextual Retrieval, and LLM-generated DSL summaries.
- Optimized the retrieval engine using metadata filtering, hybrid search (dense + BM25), parent-child linking, and reranking.
- Benchmarked configurations (chunk sizes, embedding models, fusion strategies, LLM) to double retrieval efficiency.
- Engineered a Model Context Protocol (FastMCP) server exposing 2 retrieval tools and 3 instructional prompts for LLM agents.
- Deployed into the official enterprise AI platform using Docker, Qdrant, MongoDB, and Ollama.

Human-Robot Interaction Developer — Air France Industries, Orly, France Sep 2025 – Jan 2026
Corporate Academic Project

- Deployed autonomous visitor onboarding and multimodal interaction flows on the Pepper humanoid robot by engineering native Android applications (Kotlin, SoftBank QiSDK).
- Reverse-engineered legacy robot architecture and internal systems to build a real-time engagement pipeline combining sensor-based presence detection, voice commands, and tablet UI.
- Modernized the robot's dialogue pipeline by integrating Mistral LLM APIs, upgrading basic command handling into dynamic, context-aware conversational flows.

Backend & Cloud Developer — EPF Projets, Cachan, France Jul 2025 – Oct 2025
Freelance

- Engineered the backend for "Carbon Track", a carbon-footprint mobile application, building REST APIs (Node.js, Express) connected to ADEME and Open Food Facts environmental databases.
- Designed a hybrid database & serverless backend combining Firebase (Cloud Functions, Firestore, Auth) with local SQLite for offline data persistence.
- Automated mobile build and release pipelines using the Expo/EAS CI/CD stack to test, package, and deploy the native application onto the Google Play Store.

Technical Skills

- **Programming:** Python, Java, SQL, Bash
- **AI Frameworks:** PyTorch, Hugging Face, Gymnasium (OpenAI Gym), LangChain, LlamaIndex
- **Transformer & LLM Architectures:** Attention Mechanisms, KV-Caching, Fine-Tuning (SFT, LoRA)
- **Reinforcement Learning:** Value-Based (Q-Learning, DQN), Policy Gradients & Actor-Critic (A2C, PPO), Reward Shaping
- **Generative AI:** AI Agents, MCP, Advanced RAG (Dense, BM25, SPLADE, GraphRAG, Reranking), Vector DBs (Qdrant), HNSW
- **Infrastructure & Tools:** Linux, Docker, Git, CI/CD, Cloud (AWS, GCP, Azure)
- **Languages:** French (Native), English (Professional, TOEIC 925), Spanish (Intermediate), Korean (Basic, SKKU Coursework)